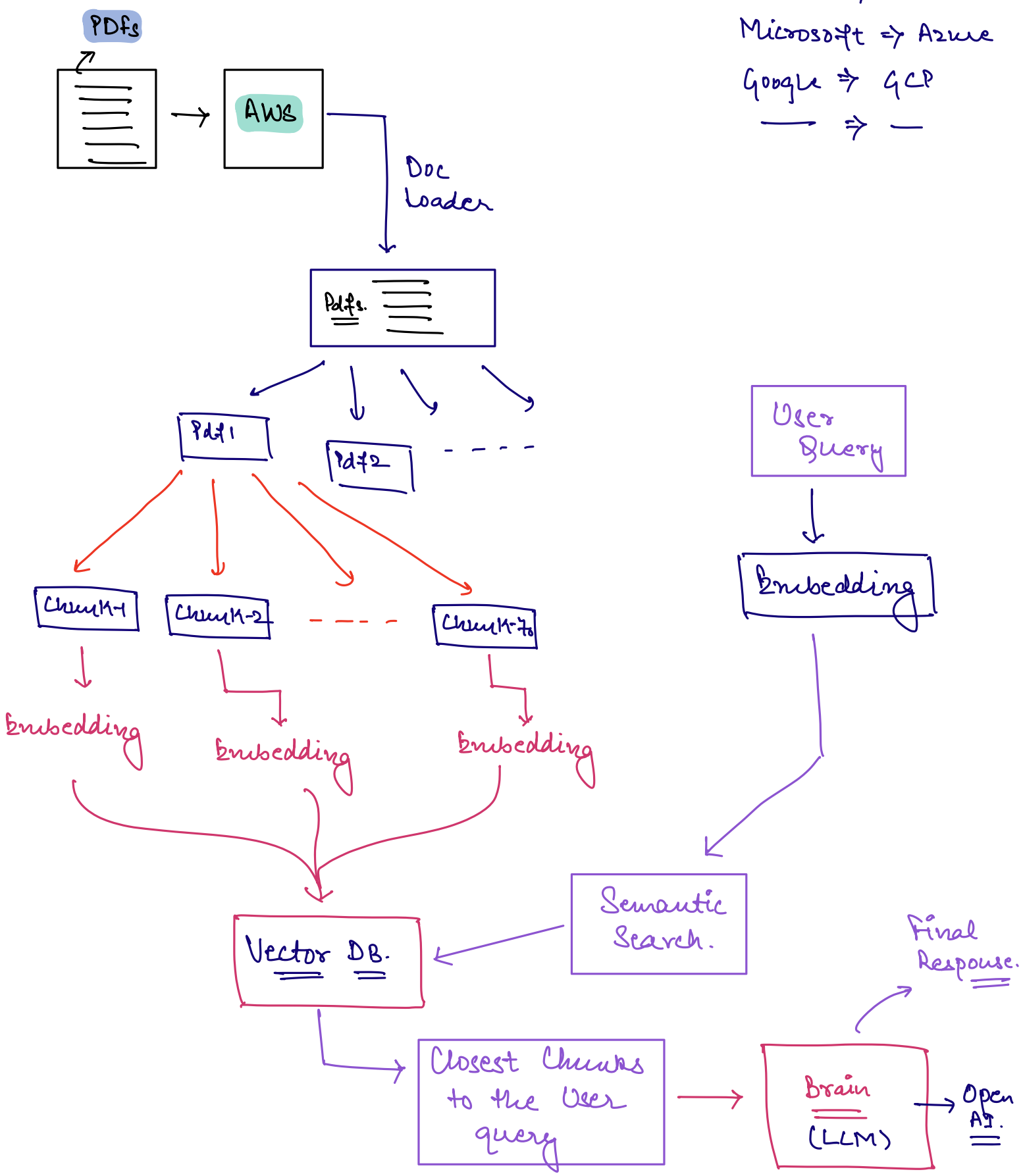
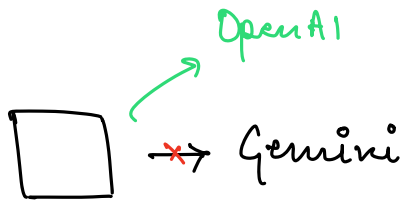
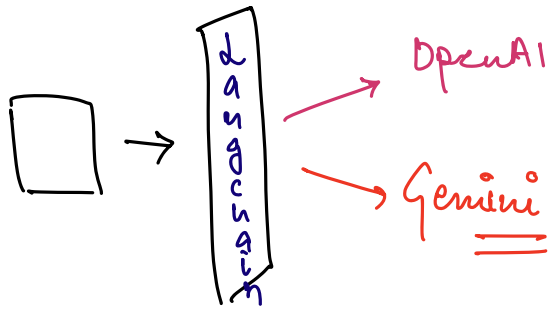
LangChain.

LLM - Based Appⁿ.

Amazon ⇒ AWS
Microsoft ⇒ Azure
Google ⇒ GCP
— ⇒ —







langchain is a framework that helps us to build LLM powered applications by combining models, prompts & tools etc that can also decide when to use tool & when to answer directly.

→ Model ≡ LLM

→ Prompt | Message ⇒ Instruction to LLM.

→ *Chain ⇒ Series | Pipeline

→ Tool ⇒ fun^n call

→ Memory | State ⇒ What the app^n remembers during or across the conversations.

LangChain Components.

1) Model.
   - LLM
   - Embedding Model.

2) Message | Prompt

3) Prompt Template.
   - ↳ Reusable prompts with placeholders.

4) Chain.
   - ↳ A fixed sequence of steps.

5) Tool
   - ↳ functions that we call for some task.
   - ↳ Model decides what tool to call & what arguments.
   - ↳ Tools can be used to fetch real-time data, call APIs, query databases etc.

6) Agents

- Agents are helpful when the system must decide autonomously what to do next?
- Agents can call one or more tools to take some action.

7) Memory | State.

- Short-Term
- Long-Term

Output Parser.

↳ Converts raw model output into the format that our application expects.

Conversation History.

↳ Older conversations & messages inside each conversation.

Agents.

→ ReAct Agents.

↳ Reason + Action

Tools.

→ External tools

→ Custom tools